CELLULAR DIVISION

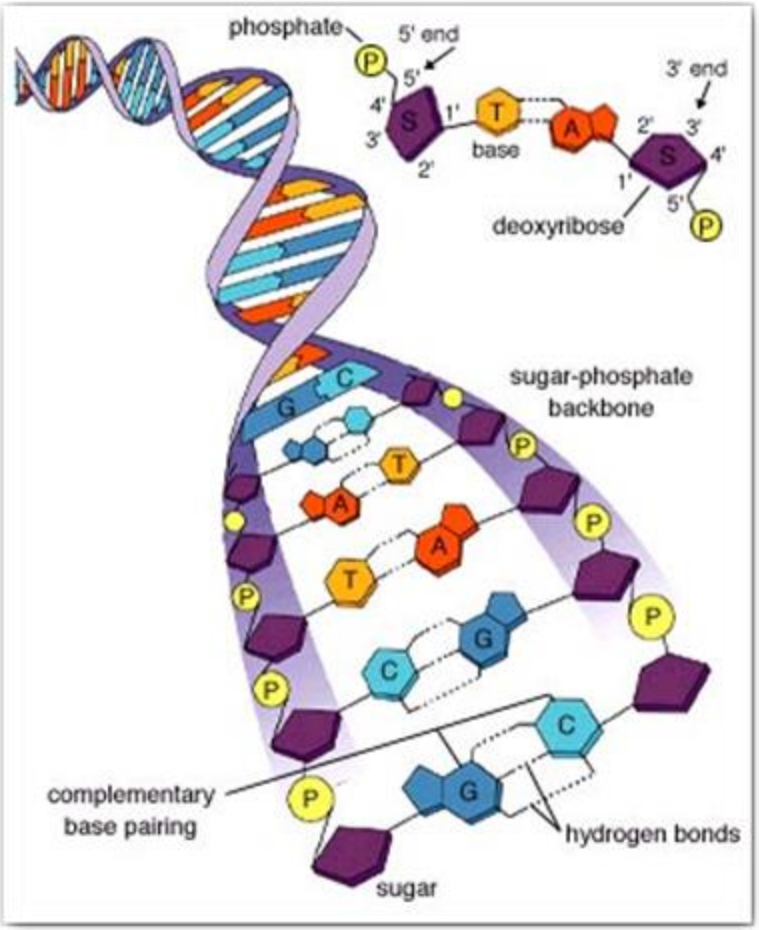
Lec 13, 14

Cell Division

- All cells are derived from pre-existing cells
- New cells are produced for growth and to replace damaged or old cells
- Differs in prokaryotes (bacteria) and eukaryotes (protists, fungi, plants, & animals)

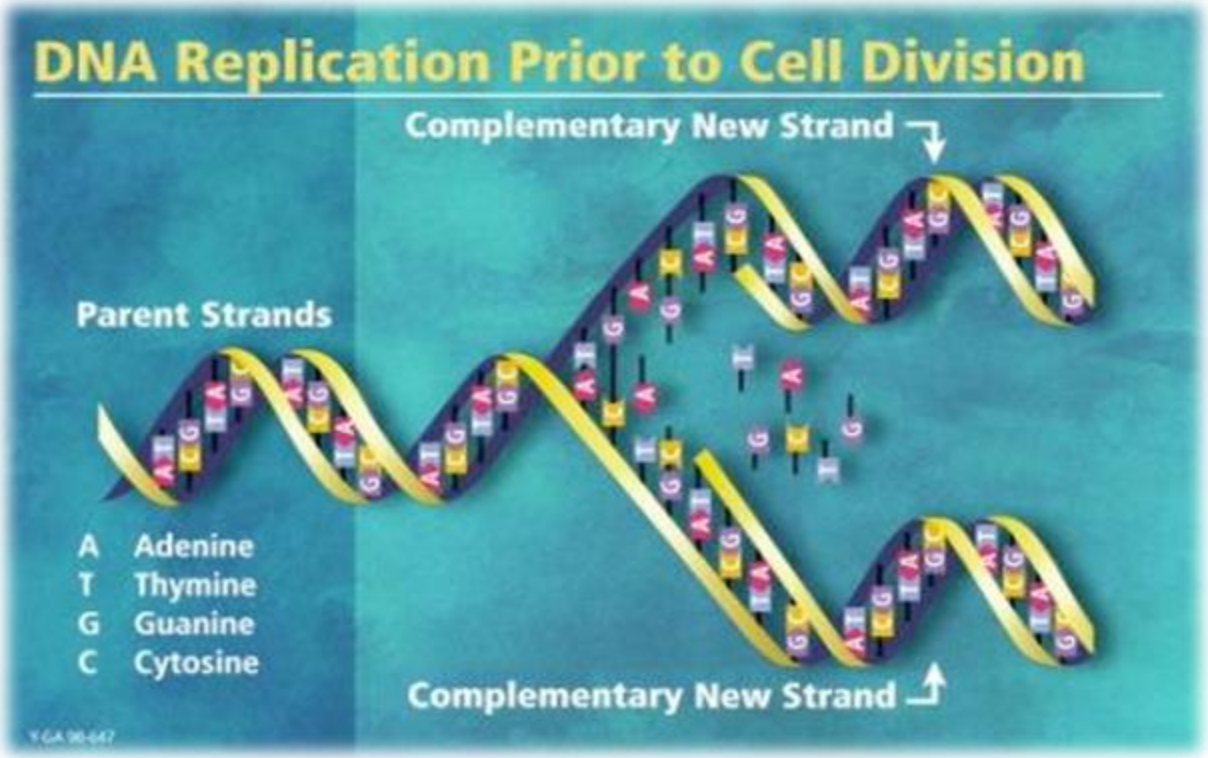
Keeping Cells Identical

- The instructions for making cell parts are encoded in the DNA, so each new cell must get a complete set of the DNA molecules



DNA Replication

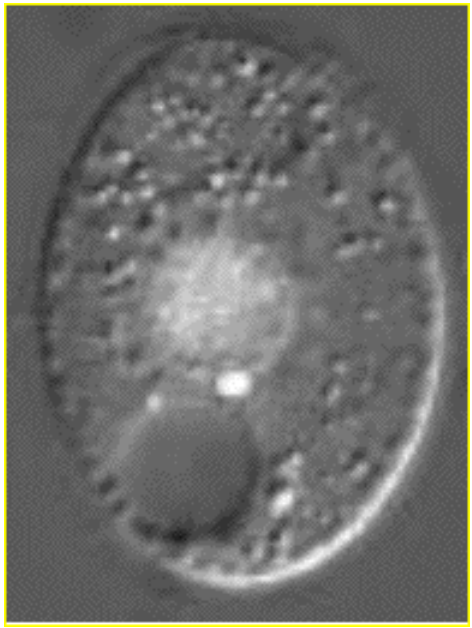
- DNA must be copied or replicated before cell division
- Each new cell will then have an identical copy of the DNA



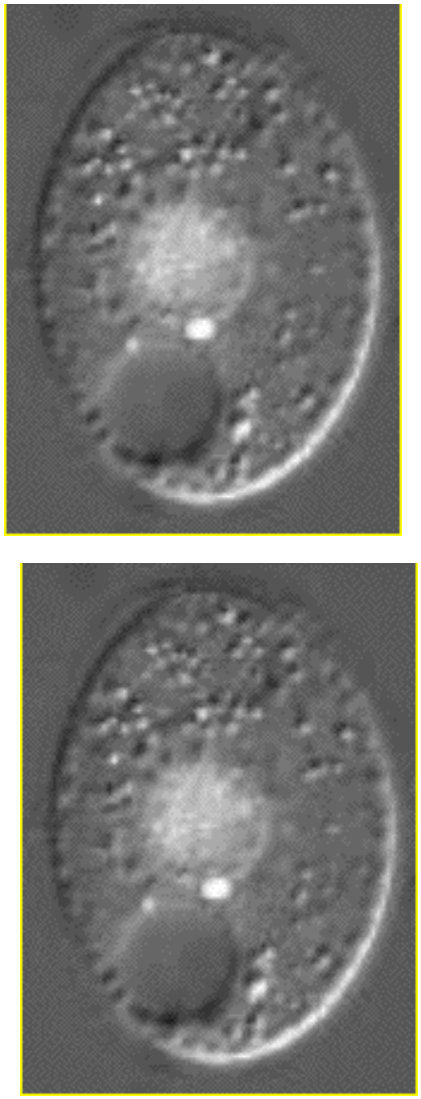
Original DNA strand

Two new, identical DNA strands

Identical Daughter Cells



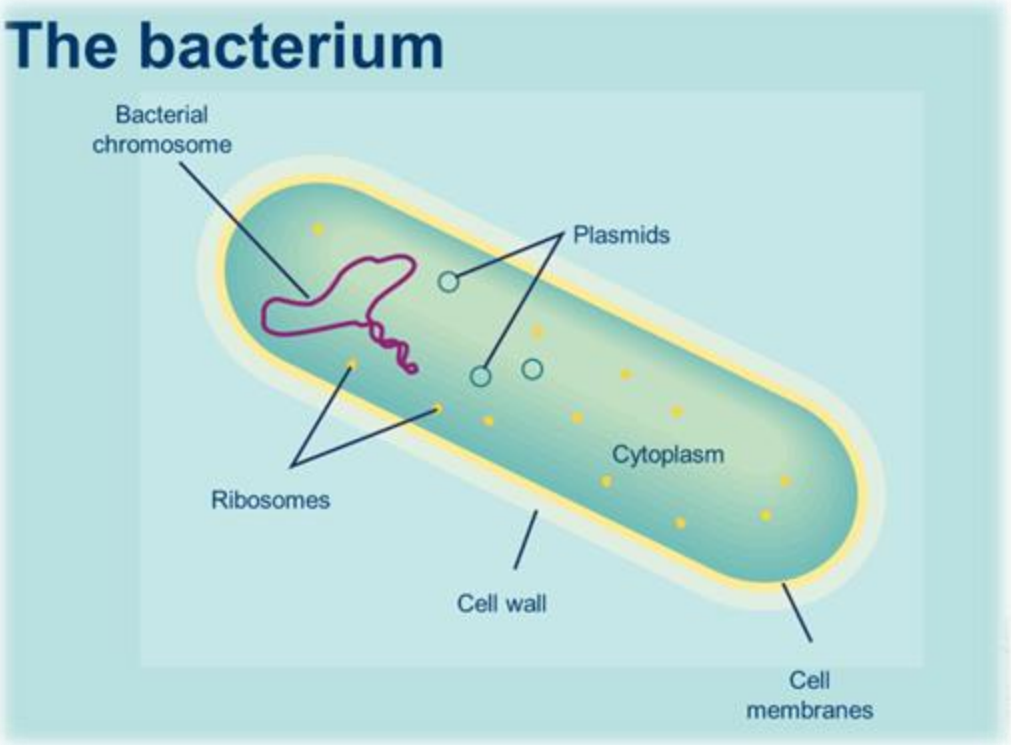
Parent Cell



Two identical daughter cells

Prokaryotic Chromosome

- The DNA of prokaryotes (bacteria) is one, circular chromosome attached to the inside of the cell membrane

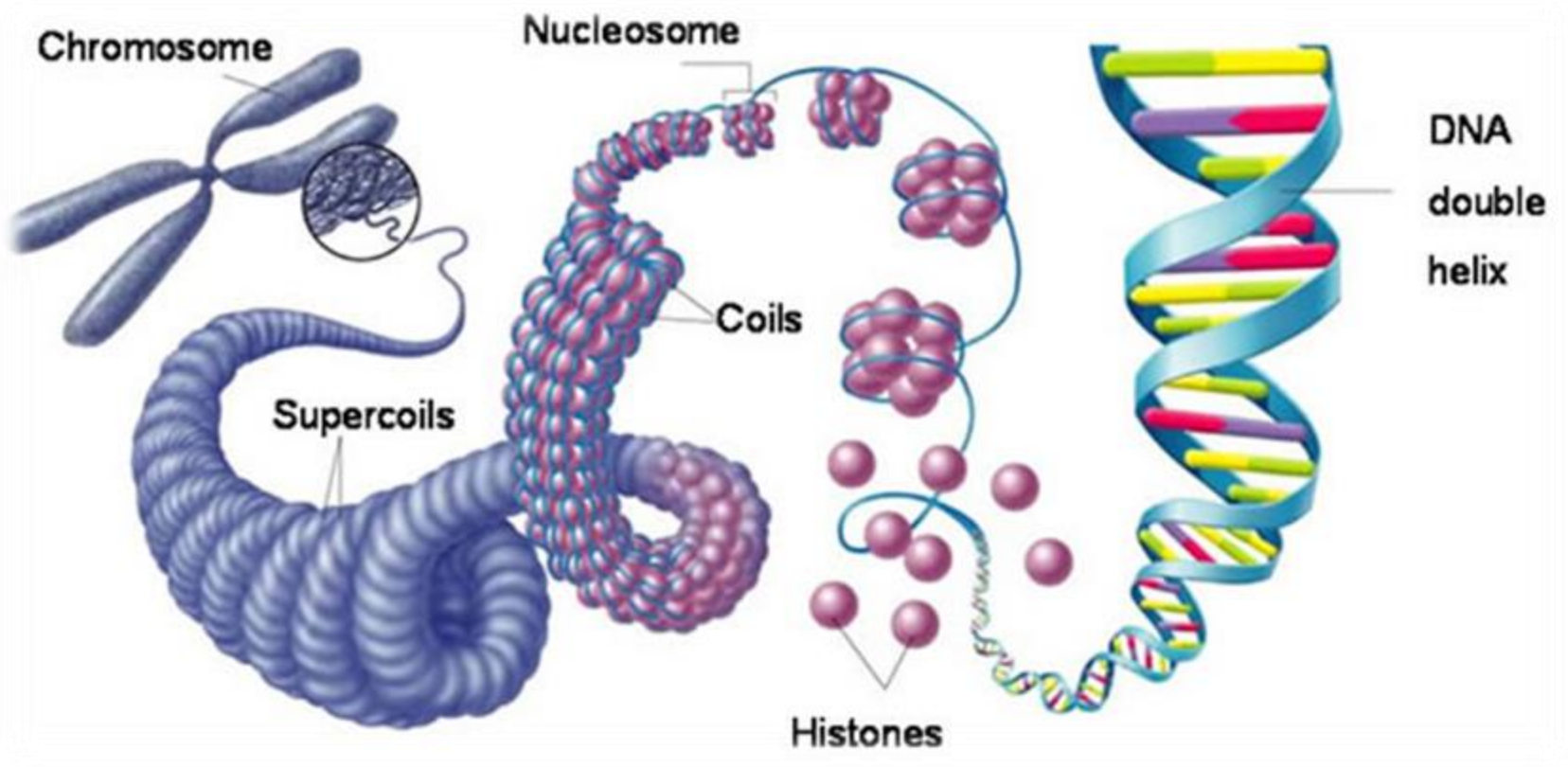


Eukaryotic Chromosomes

- All eukaryotic cells store genetic information in chromosomes
- Most eukaryotes have between 10 and 50 chromosomes in their body cells
- Human body cells have 46 chromosomes or 23 identical pairs

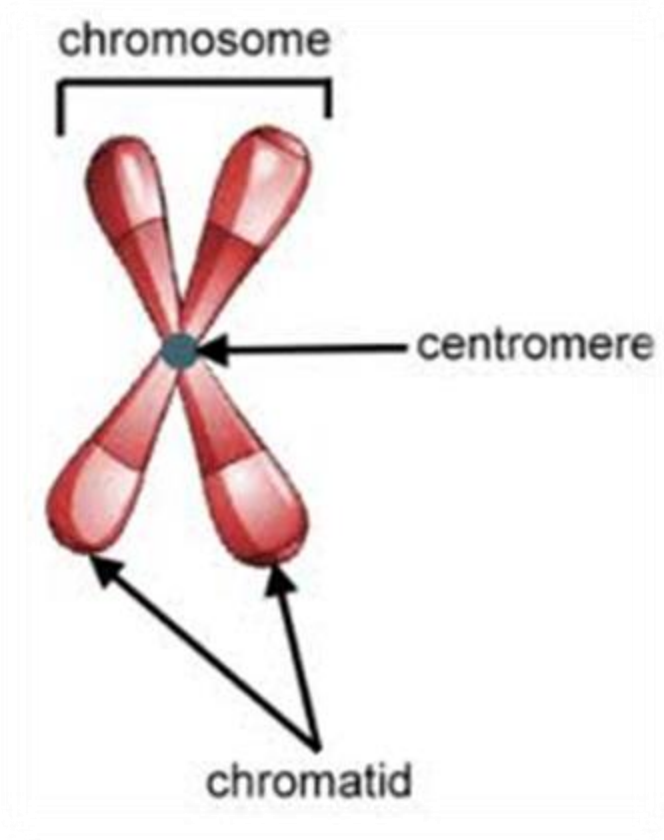
Compacting DNA into Chromosomes

- DNA is tightly coiled around proteins called histones



Chromosomes in Dividing Cells

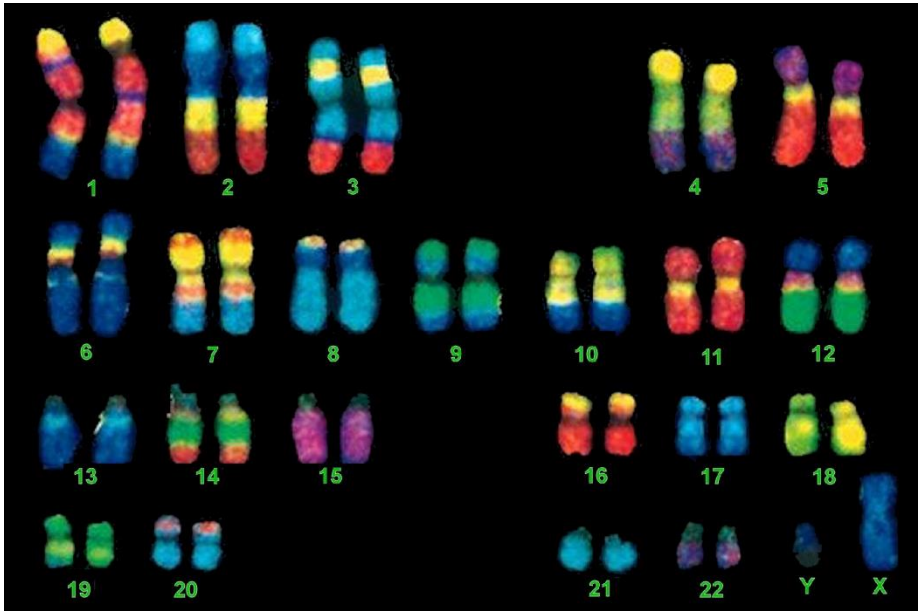
- Duplicated chromosomes are called chromatids & are held together by the centromere



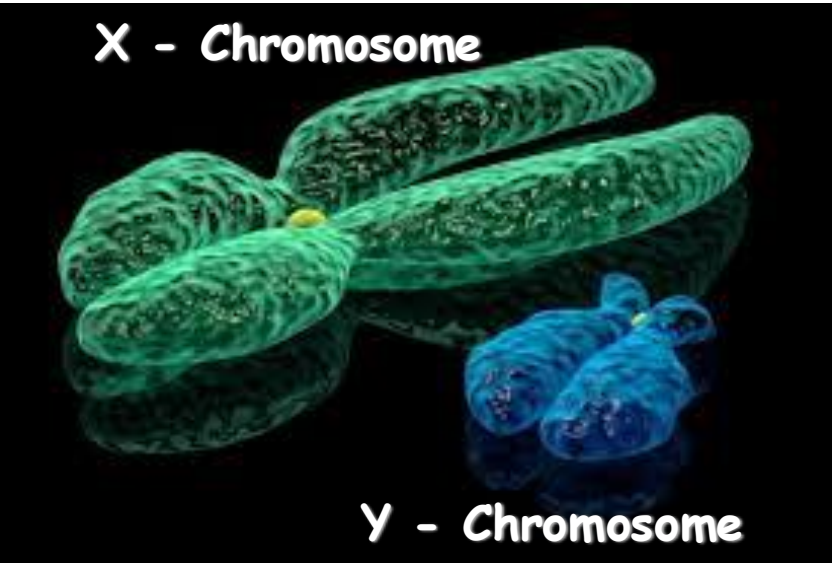
Called Sister Chromatids

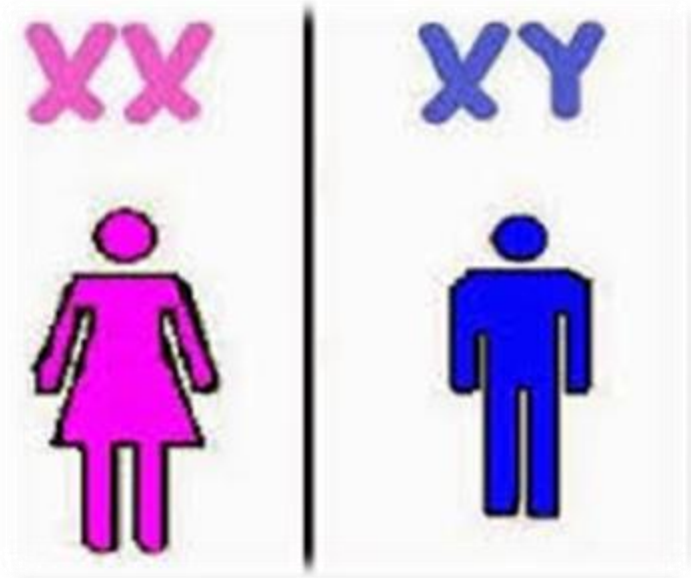
Karyotype

- A picture of the chromosomes from a human cell arranged in pairs by size
- First 22 pairs are called autosomes
- Last pair are the sex chromosomes
- XX female or XY male



Boy or Girl?





Aging

All cells die after a certain number of divisions (programmed cell death-"apoptosis"). At any given time some cells are dividing and some cells are dying

Childhood Cell division > cell death

Adulthood Cell division = cell death

Aging Cell division < cell death

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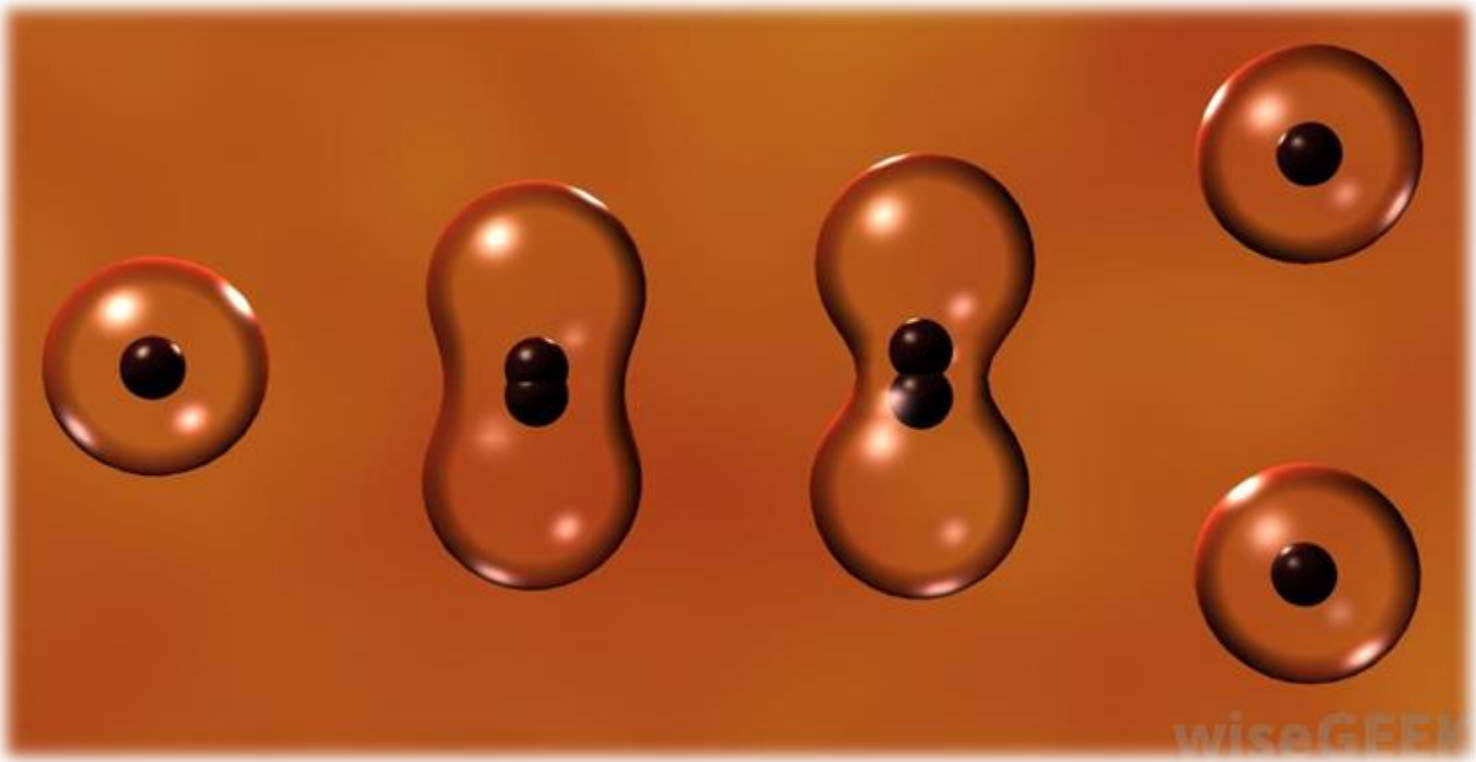
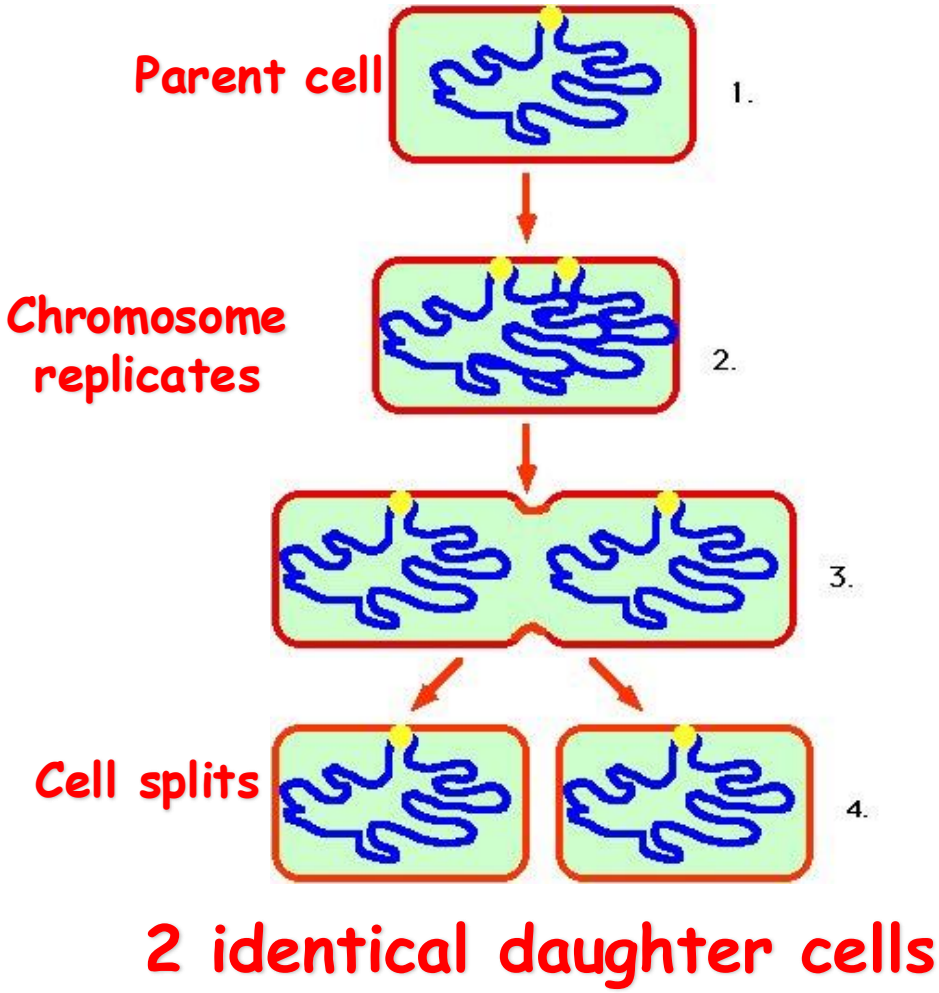
Cell Reproduction

Cell Division in Prokaryotes



Cell Division in Prokaryotes

- Prokaryotes such as bacteria divide into 2 identical cells by the process of binary fission
- Single chromosome makes a copy of itself
- Cell wall forms between the chromosomes dividing the cell



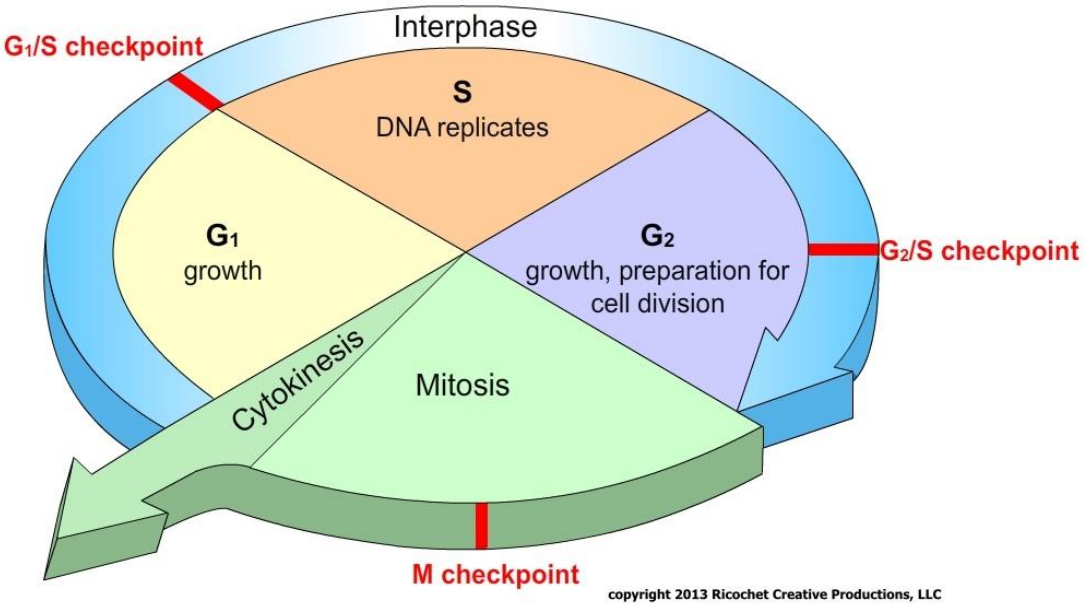
Prokaryotic Cell Undergoing Binary Fission

THE CELL CYCLE

Five Phases of the Cell Cycle

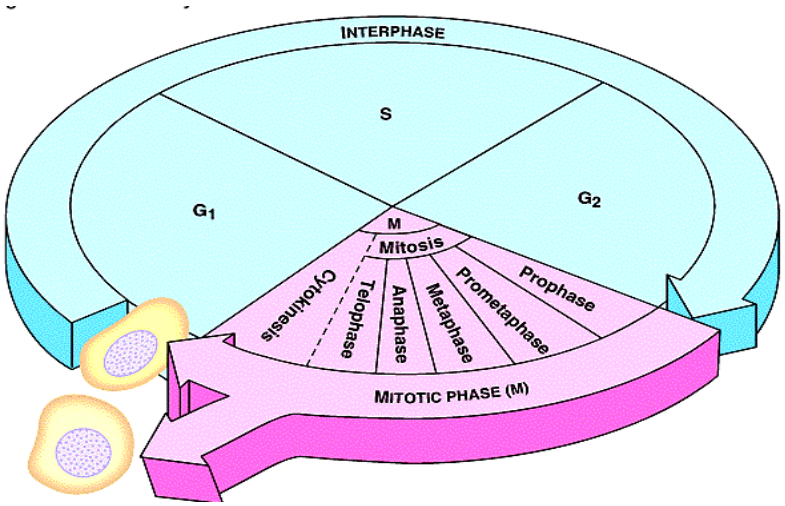
- G1 - primary growth phase
- S – synthesis; DNA replicated
- G2 - secondary growth phase
  - collectively these 3 stages are called interphase
- M - mitosis
- C - cytokinesis

Interphase ~ 90% of the time.



Mitosis

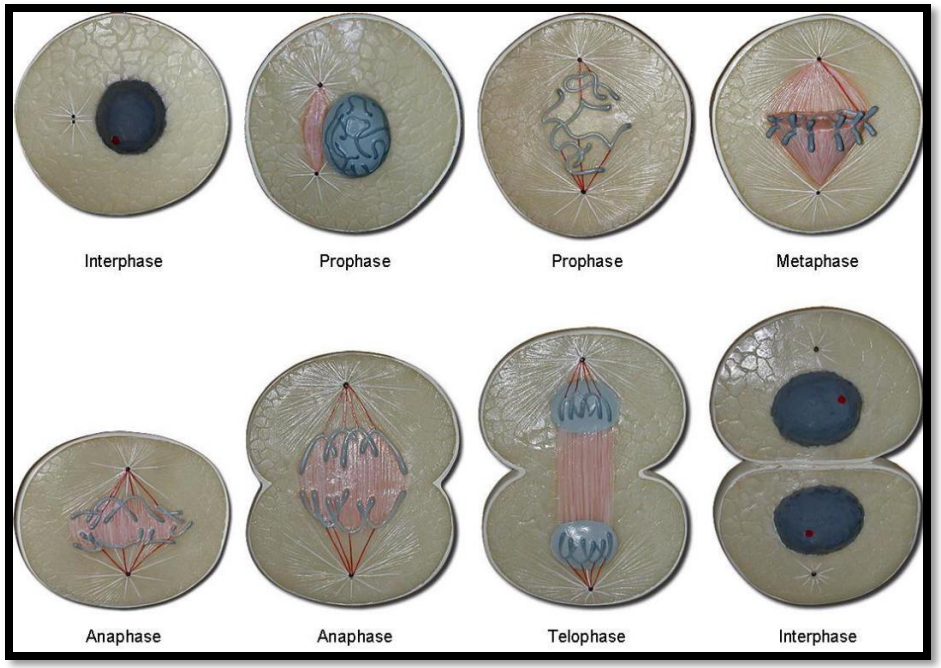
- Division of <u>somatic</u> cells (body cells)
  - (non reproductive cells) in eukaryotic organisms
  - A single cell divides into two identical daughter cells (cellular reproduction)
  - Maintains chromosome ploidy of cell



- Ploidy – refers to the <u>number</u> <u>of</u> <u>pairs</u> of chromosomes in cells
  - <u>Haploid</u> – one copy of each chromosome – designated as "<u>n</u>"
  - <u>Diploid</u> – two copies (pair) of each chromosome – designated as "<u>2n</u>"

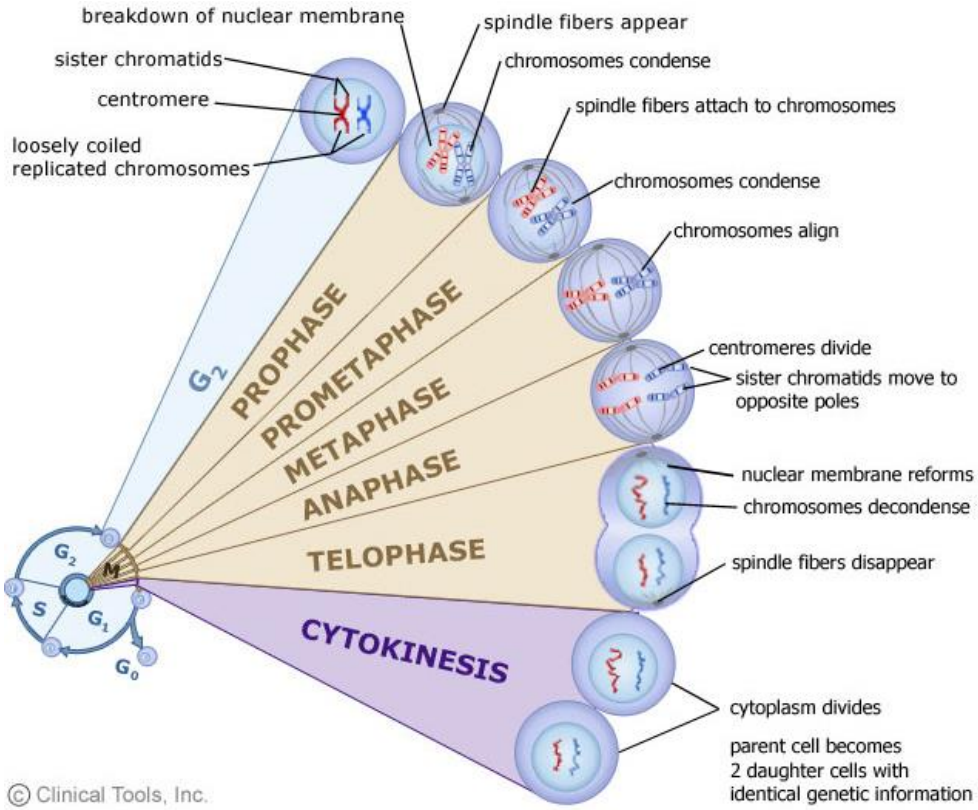
Mitosis

- Division of the nucleus
- Also called karyokinesis
- Only occurs in eukaryotes
- Has four stages
- Doesn't occur in some cells such as brain cells



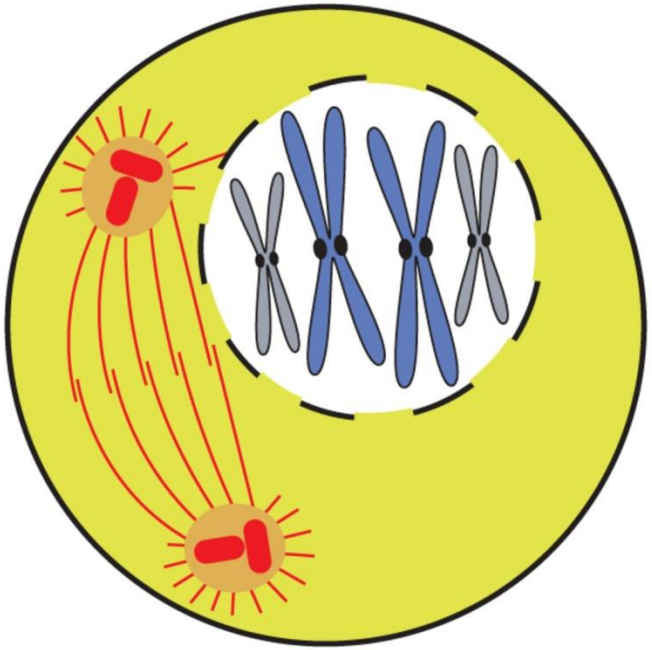
Four Mitotic Stages

- Prophase
- Metaphase
- Anaphase
- Telophase



Prophase

- Chromatin condenses (coils) into chromosomes.
- Sister chromatids joined by centromere.
- Nuclear membrane dissolves.
- Centrioles divide and move to opposite poles forming spindle between them.

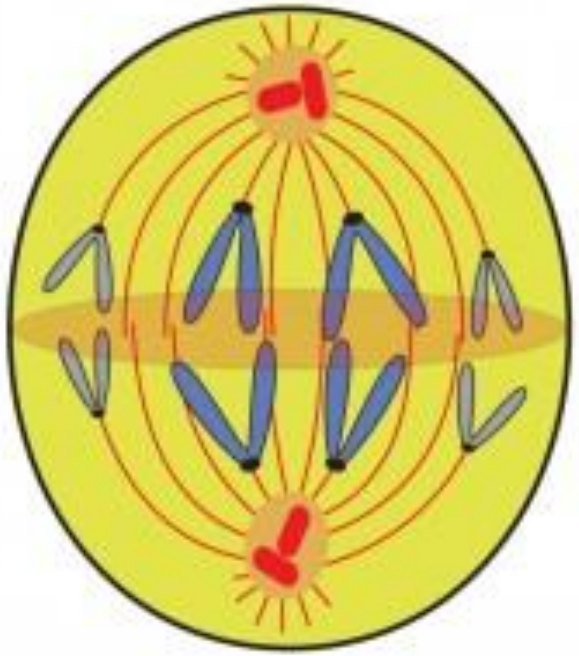


Metaphase…

- Sister chromatids line up on metaphase plate.
  - Centromeres lock on to spindle fibre

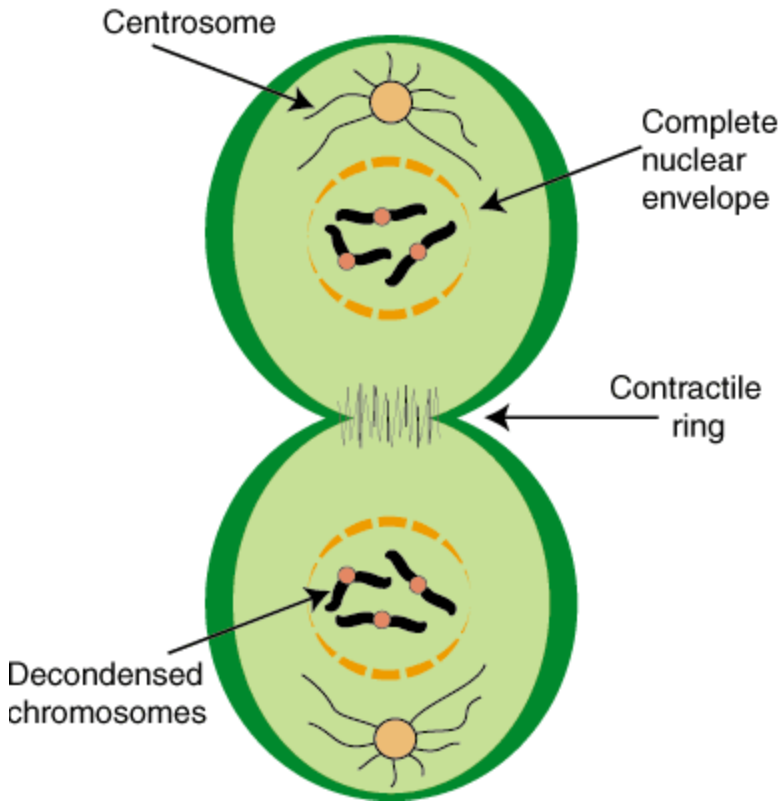
Anaphase…

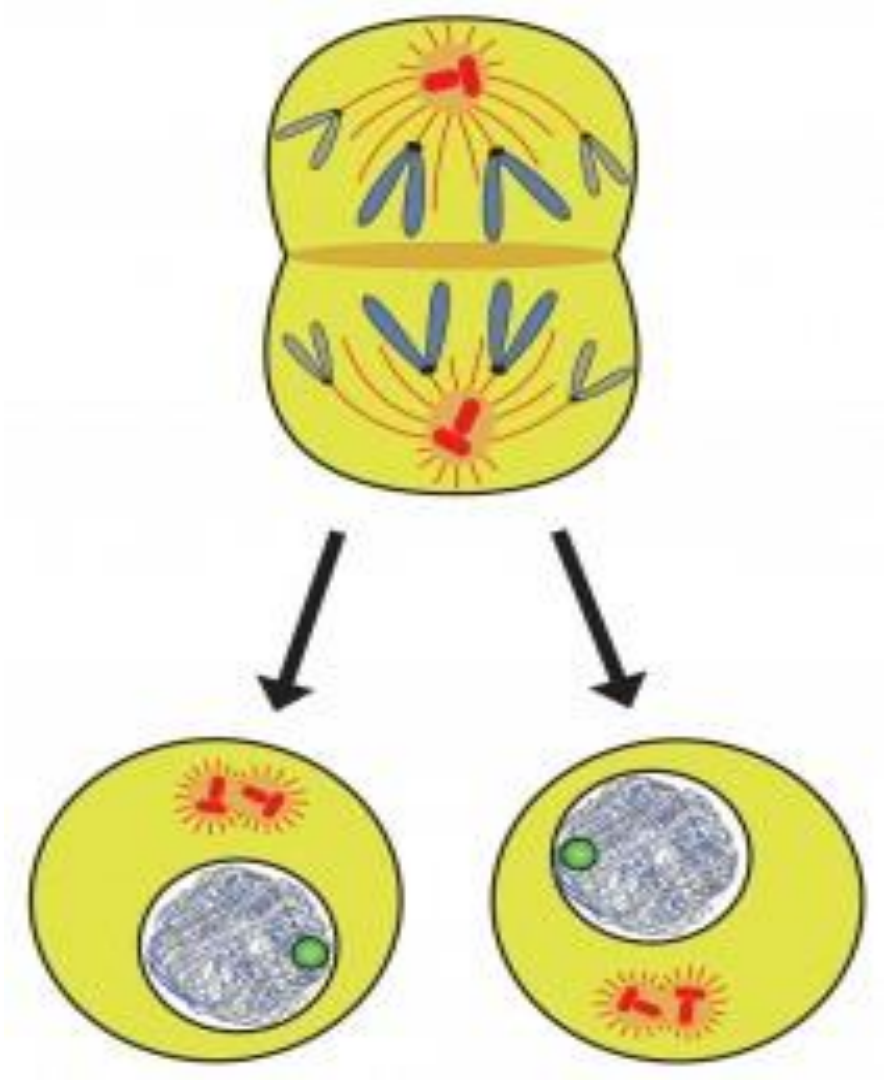
- Centromeres divide.
- Spindle fibres contract pulling sister chromatids apart to poles



Telophase…

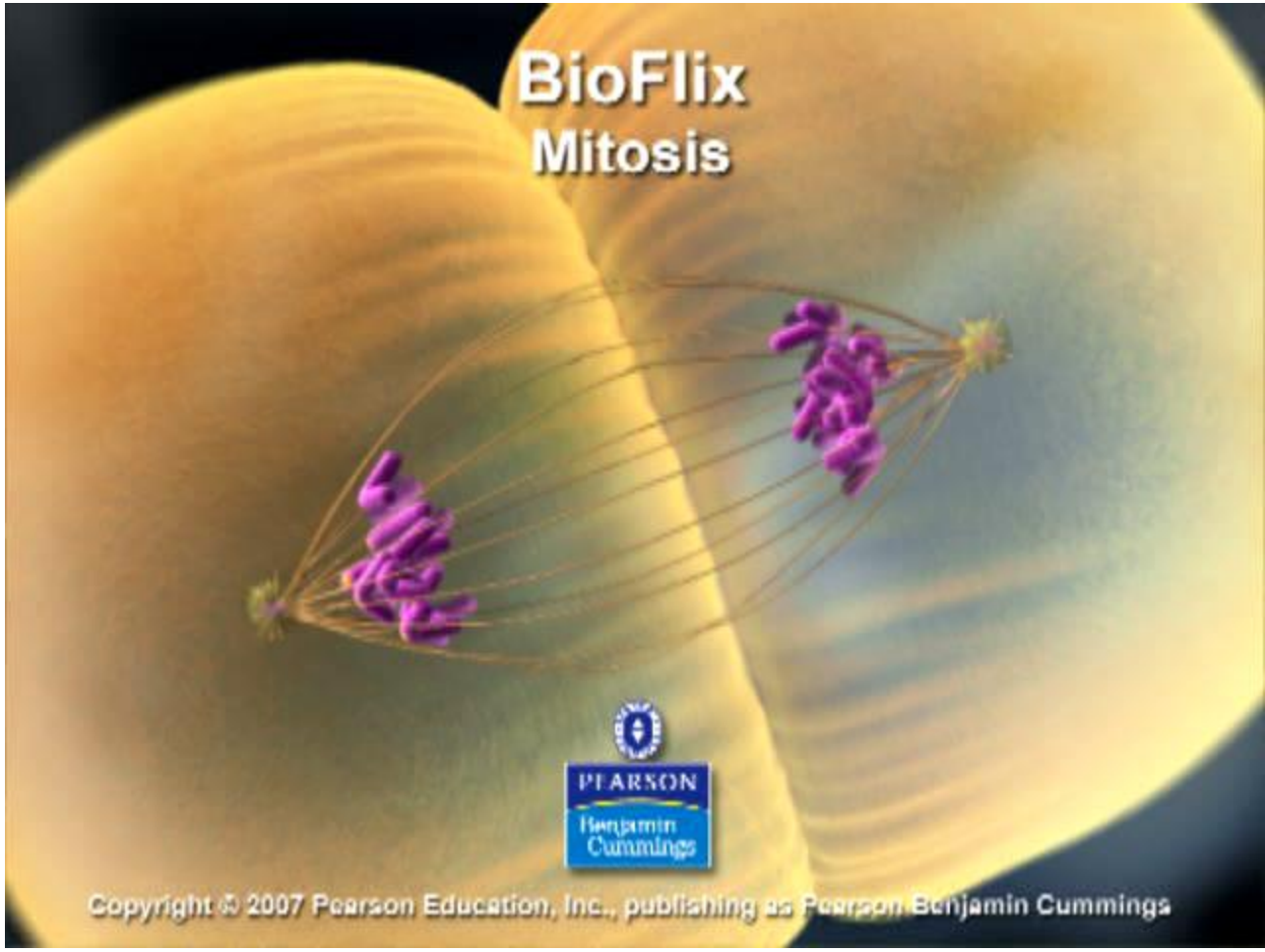
- New nuclear membranes form around new nuclei





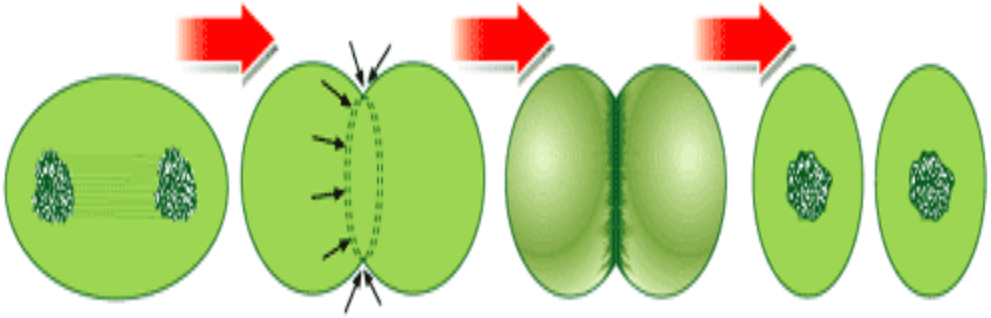
Mitosis

1. Prophase
2. Metaphase
3. Anaphase
4. Telophase



CYTO KINESIS – Cytoplasm splits into 2 cells.

-Animal cells: *Cleavage furrow* forms from outside in.

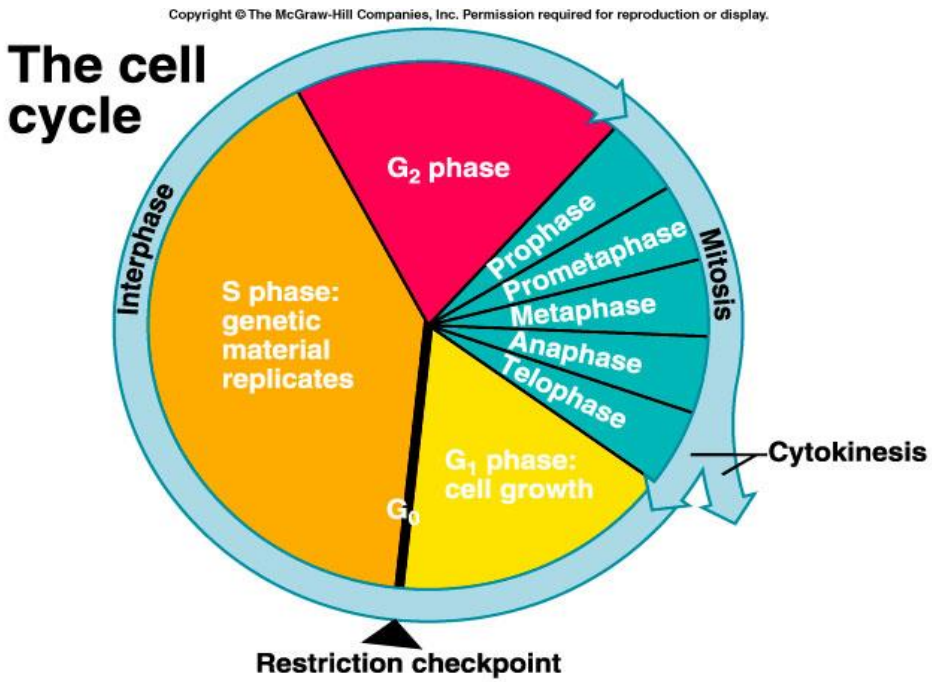


Cell cycle

Cell now returns to interphase .

The chromosomes uncoil back into chromatin.

The whole cell cycle starts over again…..



Daughter Cells of Mitosis

- Have the same number of chromosomes as each other and as the parent cell from which they were formed
- Identical to each other, but smaller than parent cell
- Must grow in size to become mature cells (G1 of Interphase)

Animation on cell cycle and mitosis

- https://www.youtube.com/watch?v=woD6zvp-4E8

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Meiosis

Formation of Gametes
(Eggs & Sperm)

Facts About Meiosis

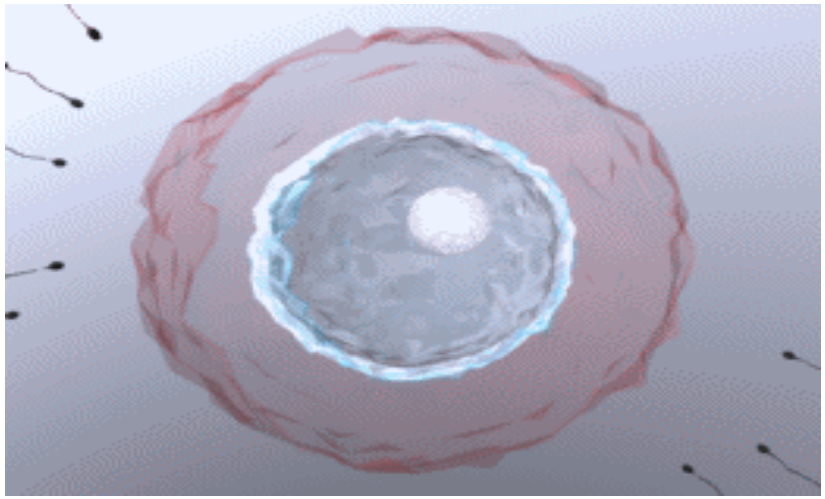
- Preceded by interphase which includes chromosome replication
- Two meiotic divisions --- Meiosis I and Meiosis II
- Called Reduction- division
- Original cell is diploid (2n)
- Four daughter cells produced that are monoploid (1n)

Facts About Meiosis

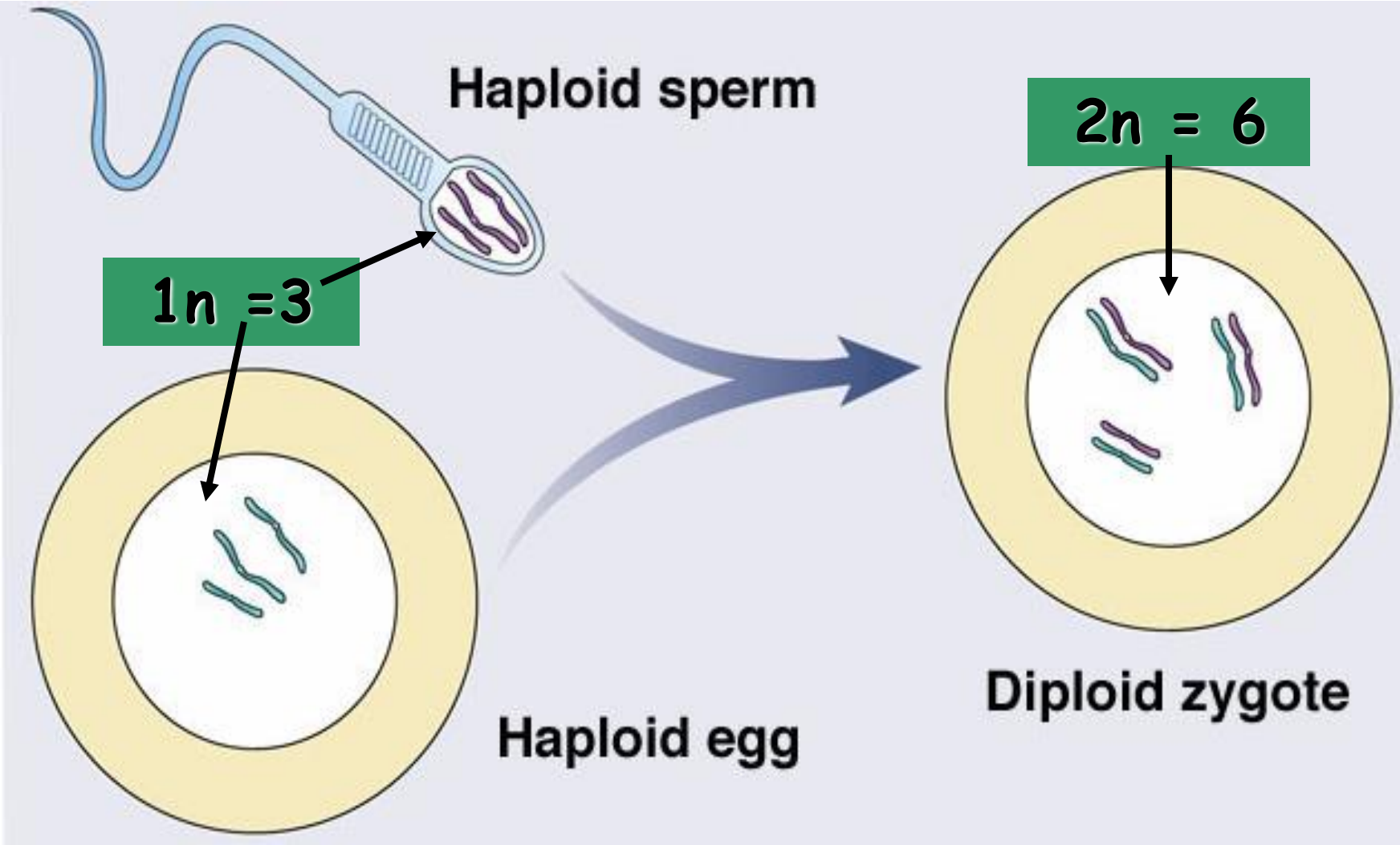
- Daughter cells contain half the number of chromosomes as the original cell
- Produces gametes (eggs & sperm)
- Occurs in the testes in males (Spermatogenesis)
- Occurs in the ovaries in females (Oogenesis)

Why Do we Need Meiosis?

- It is the fundamental basis of sexual reproduction
- Two haploid (1n) gametes are brought together through fertilization to form a diploid (2n) zygote

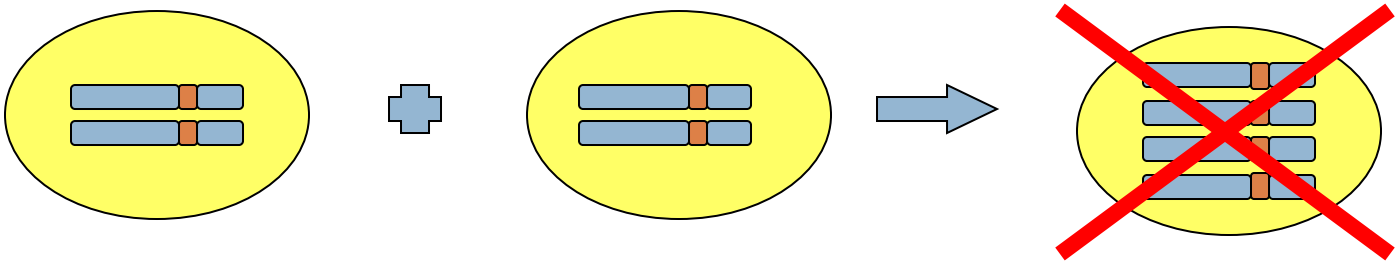


Fertilization – "Putting it all together"

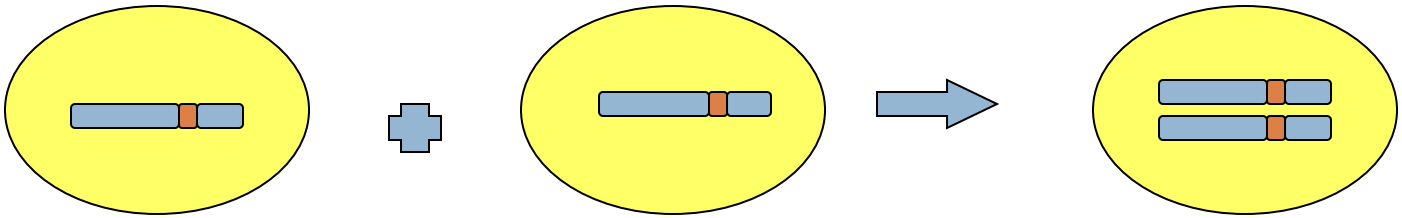


Meiosis Forms Haploid Gametes

- Meiosis must reduce the chromosome number by half
- Fertilization then restores the 2n number



too much!



The right number!

Process of Meiosis

- Meiosis animation

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Comparing Mitosis and Meiosis

Comparison of Divisions

	Mitosis	Meiosis
Number of divisions	1	2
Number of daughter cells	2	4
Genetically identical?	Yes	No
Chromosome #	Same as parent	Half of parent
Where	Somatic cells	Germ cells
When	Throughout life	At sexual maturity
Role	Growth and repair	Sexual reproduction

Types of Cell Reproduction

Asexual reproduction:
Involves a single cell dividing to make 2 new, identical daughter cells

- Binary fission and budding are examples of asexual reproduction

Sexual reproduction
Involves two cells (egg & sperm) joining to make a new cell (zygote) that is NOT identical to the original cells.

- Meiosis is an example

Asexual reproduction:

Binary Fission:

- Binary fission is the simplest form and involves the division of a single organism into two complete organisms, each identical to the other and to the parent.
- Fission is common among unicellular organisms such as bacteria, many protists, some algae such as Spirogyra and Euglena, as well as a few higher organisms such as flatworms and certain species of polychaete worms.

Asexual reproduction:

Regeneration:

- A similar form of asexual reproduction is regeneration, in which an entire organism may be generated from a part of its parent.

- The term regeneration normally refers to **re-growth** of missing, or damaged body parts in higher organisms, but whole body regeneration occurs in Hydra, starfish, and many plants.

Asexual reproduction:

Asexual Spore:

- Spores are another form of asexual reproduction and are common among bacteria, protists, and fungi.

- Spores are DNA-containing capsules capable of sprouting into new organisms; unlike most seeds, spores are produced without sexual union of gametes, that is, reproductive cells.

Asexual reproduction:

Budding:

- Budding is another method of asexual reproduction in which a group of self-supportive cells sprouts from and then detaches from the parent organism.
- Unlike eggs or spores, buds are multicellular and usually contain more than one cell layer.
- Hydra and sea squirts reproduce by budding.

Asexual reproduction:

Vegetative Reproduction:

- Vegetative reproduction is common among plants and consists of certain parts that grow out from a main parent plant and eventually root and sprout to form new, independent plants.
- Examples are the runners of strawberries, the tubers of potatoes, and the bulbs of onions.

Parthenogenesis:

- Parthenogenesis is an important means of asexual reproduction in which new individuals are formed from unfertilized eggs.
- It occurs in some insects, amphibians, reptiles, and birds and in some species of plants.